A substrate recursion principle for biological information, with empirical anchoring through a templating-mode taxonomy

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We state a substrate-level principle specifying when a physical substrate-operation pair can support open-ended information-generating recursion. The principle separates four structural conditions on the substrate-operation pair — positive entropy per recognized site, unboundedly many independently addressable non-recurrent sites, same-operation recursion (possibly up to reversible involution), and heritable local drift in the recognized alphabet — from a fifth scope-defining condition that the operation be intrinsic physicochemistry rather than externally specified optimization. The recursion has no privileged starting configuration: every template is the drifted product of prior recursions, and the substrate-level analysis treats no configuration as canonical. The principle is structural; it constrains substrates, not populations, and the population-level phenomena standardly described as inheritance, selection, error catastrophe, and evolutionary plateau are derived as corollaries. The primary theorem proves necessity: within the scope-defined domain, any substrate-operation pair failing one of (R1)–(R4) cannot support open-ended biological information. A first corollary recovers a six-mode templating taxonomy as the empirical inventory of biological substrates that satisfy or partially satisfy the principle, with each mode's bounded-heredity behavior explained by the specific condition it fails. A second corollary states that mechanisms failing same-operation recursion or heritable local drift at the individual level are structurally capped at mechanism-specific ceilings under finite-population dynamics; mechanisms satisfying both are not structurally barred from cumulative climbing on tested fitness landscapes. We test this in pre-specified simulations spanning M0–M4 inheritance mechanisms: only individual-level copying with heritable local drift (the principle satisfied at M4) reaches near-target fitness in the tested model class, while M0–M3 plateau at ceilings consistent with their respective condition failures. A descriptor-relative diagnostic apparatus (population mutual information, channel mutual information against a stated comparison ensemble, per-base mechanism fidelity), with bulk-matched and structure-scrambled controls, operationalizes the corollaries and is validated on the Drt3 system: WT Drt3b is correctly classified as cyclic-conformational templating; the E26Q point mutant matches the framework's analytical prediction (marginal G fraction 0.10, observed 0.1016); AbiK same-fold non-templating is excluded; Drt3a Mode 1 is separated from Drt3b Mode 3 by 1.0 bits of channel mutual information against the stated five-channel ensemble. Family-level apparatus application across 1,232 Drt3b homologs confirms in-envelope behavior for 33 secondary-residue clades and a 17-fold elevation in dG misincorporation at the A-state for 6 E26→D primary-gate clades, the framework-predicted shift for selectivity-gate degradation. Two distinct universal-gate classes generate single-experiment falsification targets: an architectural gate (R253) whose substitution predicts cycle-architecture collapse, and a selectivity gate (G248) whose substitution predicts intact cycle with shifted product composition.

POPULAR SUMMARY

A bacterial defense protein called Drt3b, recently characterized in *E. coli*, synthesizes DNA with a strict alternating ACAC pattern without copying any template. It generates the pattern through cyclic conformational dynamics of two amino-acid residues that flip between configurations. This is biology making sequence without sequence, and it does not fit the textbook picture of inheritance.

We give a structural specification of what physical systems can support open-ended evolution. Four conditions on a substrate-operation pair are jointly necessary: recognition of multiple states at each site, unbounded site count, operation applicability to its own products, and drift in the recognized alphabet. Drt3b fails the second: its cycle has two phases regardless of product length. The framework classifies all six known biological templating mechanisms as instances or partial instances of the same specification and generates single-experiment falsification targets at universal gate residues.

I. INTRODUCTION

Biology runs on a small number of physically distinct mechanisms for copying information from one generation to the next. DNA replication and ribosomal translation are the textbook cases. Prion propagation and amyloid templating add a second physical class — a conformational state copied across protein molecules [1]. Modular biosynthetic assemblies (non-ribosomal peptide syn-

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thetases, polyketide synthases) compose a third [2]. The eukaryotic ciliate cortex constitutes a fourth: cortical patterns of basal-body orientation are inherited at division through the action of the existing cortex on newly assembled units, with grafted inverted patches propagating as inverted for hundreds of generations without genomic change [3, 4]. The recent characterization of bacterial defense-related templating systems — the DRT family [5–7] — adds at least one more: Drt3b synthesizes alternating poly(AC) DNA without using a nucleic acid template, generating its product through cyclic conformational dynamics of a protein active site rather than through Watson-Crick complementarity to an existing strand [5].

These mechanisms pose a question that is physical rather than purely biological: what structural conditions on a substrate-operation pair are necessary for it to support open-ended information-generating recursion? The mechanisms look mechanistically unrelated. Some support the open-ended evolution that biology requires — DNA replication does. Some do not — prion propagation cannot accumulate qualitatively new structures over evolutionary timescales. The field has no structural account of why. What does a physical substrate need to be in order to support open-ended evolution? Which mechanisms qualify, which do not, and on what principled basis is the line drawn? These questions are surprisingly absent from the current literature, despite half a century of attempts to address adjacent questions.

The attempts have approached the substrate-level question from three directions. The information-theoretic line — Eigen [8], Eigen *et al.* [9], and Szathmáry [10] — treats copying-with-variation as a process whose stability depends on per-site error rate relative to genome length, generating the quasispecies framework and the error threshold $\mu^* \approx \log \sigma / L$. This work tells us how a substrate-borne replication process can fail dynamically (above threshold, the master sequence is lost) but not what makes a physical system a substrate at all. Szathmáry [10] introduced the distinction between limited and unlimited heredity — a substrate with K possible states transmits limited heredity ($\log_2 K$ bits ceiling), while sequence-based substrates transmit unlimited heredity ($L \log_2 |\mathcal{A}|$, scaling with length). The distinction is intuitive and correct in spirit, but it is not formalized as a structural condition on substrates; it is asserted as a typological observation.

The ontological line — Hull [11], Maynard Smith [12], and the philosophical-of-biology tradition that follows — specifies what kind of entity a replicator must be (a faithful copier of itself across generations, with heritable variation in fitness). This work clarifies the abstract requirements for a unit of selection but is substrate-blind: any physical realization that meets the abstract requirements counts. The substrate-level question — which physical realizations meet the requirements, and through what structural mechanism — is not addressed.

The substrate-level line — Schrödinger [13]'s aperiodic crystal, the GARD-class compositional inheritance models [14, 15], and Pross [16]'s dynamic kinetic stability — has attempted to specify the physical substrate directly. Schrödinger's gestural answer (the chromosome is an aperiodic crystal whose information is in its sequence of distinct molecular groups) was prescient but not formalized. GARD-class models attempted a formal substrate-level alternative: a compositional set of catalytic molecules can transmit information across generations through preferential reproduction of similar compositions. Vasas *et al.* [17] showed using GARD's own machinery that compositional inheritance fails the fidelity requirements for open-ended evolution: high-fidelity compositional transmission requires a lognormal distribution of catalytic rate constants so narrow as to be biologically unrealistic. The result is a clear empirical verdict (compositional inheritance does not work), but it does not generalize: it tells us GARD fails, not what any substrate must satisfy to succeed.

What is missing from all three lines is a single statement, at the substrate level, of what physical conditions a substrate-operation pair must satisfy to support open-ended biological information. The information-theoretic work is at the wrong level (population dynamics on substrates, not the substrates). The ontological work is at the wrong level of abstraction (replicators in the abstract, substrate-blind). The substrate-level work has either gestured (Schrödinger) or been falsified (GARD). The Drt3b finding makes the absence acute: Drt3b is unambiguously biological, unambiguously templating, and unambiguously not a sequence-template system. A framework that classifies it has to be substrate-level and structural. There is no other place to put it.

We give such a framework here. The substrate recursion principle states four structural conditions on a substrate-operation pair that are jointly necessary for open-ended information-generating recursion — positive entropy per recognized site (R1), unboundedly many independently addressable non-recurrent sites (R2), same-operation recursion possibly via reversible involution (R3), and heritable local drift in the recognized alphabet (R4) — together with a fifth scope-defining condition that the operation be intrinsic physicochemistry rather than externally specified optimization (R5). The principle is structural: it states what a substrate-operation pair must be, not what populations of such substrates do under selection. Selection, fitness, error thresholds, and evolutionary plateaus are downstream consequences. We prove that any substrate within the (R5) scope failing one of (R1)–(R4) cannot support open-ended biological information.

The principle has direct empirical content through two corollaries. The first describes bounded heredity under condition failure: a substrate failing one structural condition still exhibits heredity, but the heredity is bounded in a manner determined by which condition fails. This corollary classifies biology's six known templating mechanisms as a substrate-mode taxonomy. Mode 1 (DNA replication, Drt3a, telomerase, β-sheet peptide replication) satisfies all four structural conditions within the (R5) scope. Mode 2 (ribosomal translation) fails (R3): the protein product is not ribosome-readable, and the recursion is recovered parasitically through Mode 1 acting on the gene that encodes the lookup-table machinery. Mode 3 (Drt3b, the cyclic-conformational case that motivated this work) fails (R2): the cycle has only N phases regardless of product length, capping the substrate at $\log_2 N$ bits. Mode 4 (prions, amyloids, DRT1) fails (R1) at the macro level: the recognition alphabet is the small attractor repertoire of conformers, bounded at $\log_2 K$. Mode 5 (NRPS, polyketide synthases) fails (R2) at the module count and (R3) at the substrate level. Mode 6 (the ciliate cortex) satisfies (R1) and (R2) at the surface level but fails (R3) and (R4): the surface does not autonomously self-copy, and pattern drift independent of genome drift cannot accumulate as heritable variation in the recursive sense.

The second corollary states that mechanisms failing same-operation recursion or heritable local drift at the individual level are structurally capped at mechanism-specific ceilings under finite-population dynamics. We test this in pre-specified simulations across five inheritance mechanisms (M0–M4), where M4 satisfies the structural conditions at the individual level and M0–M3 fail specific conditions. Only M4 reaches near-target fitness in the tested model class; the remaining mechanisms plateau at ceilings consistent with their respective condition failures. The corollary's positive direction is bounded: substrates satisfying (R3) and (R4) at the individual level are not structurally barred from cumulative climbing, but actual climbing depends on landscape topology and other conditions outside the principle's scope.

The framework's empirical anchoring is provided by a pre-specified diagnostic apparatus comprising population mutual information against a stated template population, channel mutual information against a stated comparison ensemble of templating channels, and per-base mechanism fidelity components. The apparatus correctly classifies Drt3b WT as cyclic-conformational templating, the E26Q gate-broken mutant with quantitative match to the published observation (marginal G fraction 0.10, observed 0.1016), AbiK same-fold non-templating as excluded, and the Drt3a-versus-Drt3b channel separation at 1.0 bits against the stated five-channel ensemble. Family-level apparatus application across 1,232 Drt3b homologs confirms in-envelope behavior for 33 secondary-residue clades and a 17-fold elevation in marginal dG fraction for 6 E26→D primary-gate clades, the framework-predicted selectivity shift. Two universal-gate classes generate single-experiment falsification targets through site-directed mutagenesis: an architectural gate (R253) whose substitution predicts cycle-architecture collapse, and a selectivity gate (G248) whose substitution predicts intact cycle with shifted product composition.

The framework's contribution sits on three legs: the principle as conceptual core, the substrate-mode taxonomy as empirical instantiation, and the diagnostic apparatus as operational instrument. The principle's substrate-essentialism carries one further entailment that we make explicit. The recursion has no privileged starting configuration. Every template is the drifted product of prior recursions, and the substrate-level analysis treats no configuration as canonical. We accordingly use the vocabulary of "drift" rather than "error" or "variation" wherever the latter would smuggle a target back in — "error" presupposes a correct answer the operation was trying for, and the principle's content is precisely that the operation has no such target. We retain "error" for established field-standard usage (Eigen's error threshold, per-position misincorporation rate ε against a defined Watson-Crick canonical product) where the technical meaning is fixed and replacement would lose precision.

II. RESULTS

A. The substrate recursion principle: structural conditions and necessity

We state the principle in three parts: an independent definition of open-ended information-generating recursion, the substrate recursion conditions, and the necessity theorem. We give each formal statement after a brief plain-language framing of what it claims, and follow each with a biological example showing what the formalism captures.

a. Definition 1: what counts as open-ended information generation. Before specifying conditions on the substrate, we have to specify what the substrate is required to support. "Open-ended information generation" is used loosely in the literature; the formal version below makes it precise as a scaling property of the configuration repertoire. The intuition: starting from any initial configuration, repeated application of the substrate's operation should be able to produce a number of distinct templatable configurations that grows exponentially with the substrate's size parameter L (sequence length, surface area, module count). Polynomial growth is not enough; bounded growth is certainly not enough. Biology's open-endedness is the empirical fact that DNA-based lineages have not exhausted the space of distinguishable genomes after four billion years.

Definition 1 (Open-ended information-generating recursion)**.** A substrate-operation pair $(\mathcal{X}, \mathcal{O})$ supports open-ended information-generating recursion if iterated application of its operation $\mathcal{O}$ generates an unbounded sequence of distinct, copyable configurations whose drift remains available as templates for future iterations. Formally: the configuration repertoire after T iterations, starting from any $X_0 \in \mathcal{X}_L$, is not bounded above by any function $f(L)$ that fails to scale as $\Theta(2^{cL})$ for some

$c > 0$.

DNA is the canonical example. A length-L DNA sequence has 4^L possible distinct configurations; iterated replication with drift accumulates configurations from this exponentially growing space without exhausting it. Drt3b is the canonical non-example: at any product length, Drt3b's cyclic dynamics have produced one of two phases at each position (A or C), and the substrate's distinguishable-state repertoire is bounded at $\log_2 2 = 1$ bit regardless of how long the product gets.

The recursion has no privileged starting configuration. Every template X_t is the drifted product of prior recursions $X_{t-1}, X_{t-2}, \ldots$, and the substrate-level analysis treats no configuration as canonical: there is no Platonic ideal sequence the recursion is trying to reach, only the actual physical templates the substrate has produced.

b. Definition 2: structural conditions on the substrate-operation pair. The substrate-operation pair $(\mathcal{X}, \mathcal{O})$ is the formal object the principle constrains. $\mathcal{X}_L$ is the space of admissible configurations of size L; $\varphi_X(X) \in \Sigma^L$ is the descriptor of X over the recognition alphabet Σ; $\mathcal{O}$ is the operation with kernel $Q_L(X' \mid X)$ producing $X' = \mathcal{O}(X)$, possibly composed with a reversible involution $\iota : \mathcal{X}_L \to \mathcal{X}_L$. The reversible involution covers the canonical case where the immediate product of $\mathcal{O}$ is the complement of the input (Watson-Crick complementarity in DNA replication produces $\bar{X}$, and the next round of replication on $\bar{X}$ recovers X); the involution permits the recursion to close even when each individual $\mathcal{O}$ application produces a complement rather than a copy.

Five conditions follow. Four are structural and are what the necessity theorem will require; the fifth is scope-defining and demarcates the domain to which the principle applies. We state them formally first, then walk through each in biological terms.

Definition 2 (Substrate recursion conditions)**.** The substrate-operation pair $(\mathcal{X}, \mathcal{O})$ satisfies the substrate recursion conditions if:

(R1) Positive entropy per recognized site.: At each addressable position i, the recognition step in $\mathcal{O}$ distinguishes at least two alternative monomer states drawn from a recognition alphabet Σ with $|\Sigma| \geq 2$. The per-site entropy under the maximum-entropy distribution on the alphabet, $h_{\text{site}} = \log_2 |\Sigma|$, is positive: $h_{\text{site}} > 0$.

(R2) Unbounded extensible position count.: The substrate has independently addressable non-recurrent sites whose count $N(L)$ grows unboundedly: $\liminf_{L\to\infty} N(L) = \infty$. Composition vectors, homogeneous concentrations, and bulk mixtures fail because they lack addressable position. Cyclic systems with finite recurrent phase repertoires fail because $N(L)$ is bounded at the cycle length.

(R3) Same-operation recursion (with allowed involution).: The output $X' = \mathcal{O}(X)$ is structurally an element of $\mathcal{X}_L$ for which $\mathcal{O}(X')$ is well-defined, possibly via a reversible involution ι such that $\mathcal{O}(\iota(X')) \in \mathcal{X}_L$. The recursion takes the form

$$X \xrightarrow{\mathcal{O}} X' \xrightarrow{\mathcal{O}} X'' \xrightarrow{\mathcal{O}} \cdots \quad \text{or} \quad X \xrightarrow{\mathcal{O}} \iota(X') \xrightarrow{\mathcal{O}} X'' \xrightarrow{\mathcal{O}} \cdots$$

The condition excludes operations whose products are not readable by the same operation.

(R4) Heritable local drift in the recognized alphabet.: The kernel $Q_L(X' \mid X)$ has nonzero probability of producing configurations $X' \neq X_{\text{ref}}(X)$ for any reference configuration, with the drift lying in the same alphabet Σ as the recognition step in (R1). Substrate content (the descriptor $\varphi_X(X)$) and substrate templating function (the role of X in $\mathcal{O}(X) \to X'$) are the same physical entity: drift in content is drift in the next iteration's template. The use of *drift* rather than *error* or *variation* reflects the substrate-essentialist commitment that no configuration is canonical; the kernel produces what its physics produces, and drift is the statistical structure of that production rather than departure from a target.

(R5) Scope: non-teleological intrinsic operation.: The kernel Q_L depends only on the physicochemistry of $(\mathcal{X}, \mathcal{O})$, not on an externally specified target configuration, loss function, or convergence criterion. (R5) is scope-defining rather than structural: it demarcates the domain in which the structural conditions (R1)–(R4) apply. Differential persistence of configurations may occur downstream as a consequence of $\mathcal{O}$'s action over a finite population, but selection acts on products, not on the operation itself. Systems whose operation is externally optimized (gradient descent on a loss function, designed catalytic cycles with specified targets) are outside the principle's scope.

The conditions in biological terms:

- **(R1)** requires that the substrate's recognition step distinguishes at least two states at each site. DNA satisfies this: at each position, the polymerase distinguishes A, T, G, and C, giving $h_{\text{site}} = 2$ bits. A system that recognizes only one state at each position — a polymerase that incorporates only T — has $h_{\text{site}} = 0$ and fails (R1) at the recognition level.
- **(R2)** requires that the count of independently addressable sites grows without bound. DNA satisfies this: the polymer can extend arbitrarily, and each new position is independent of the prior. Drt3b's cycle has only two phases regardless of product length; site count is bounded at 2, and (R2) fails.
- **(R3)** requires that the substrate's operation can be applied to its own output. DNA replication satisfies

this through the involution $X \to \bar{X}$: the daughter strand is itself a DNA strand, and a polymerase can replicate it. Ribosomal translation fails (R3): the protein product of translation is not a substrate the ribosome can re-read; recursion is recovered only through Mode 1 acting on the gene that encodes the ribosomal machinery.

- **(R4)** requires that the substrate's operation produces drift, and that the drift lies in the alphabet the operation recognizes. DNA polymerases produce drift in the nucleotide alphabet (a $\sim 10^{-9}$ per-position rate against the Watson-Crick canonical product), and the drifted product is itself recognized at the next iteration. A perfect copier (zero drift) fails (R4); a system that produces drift in a different alphabet (chemical modifications that the polymerase does not read) also fails (R4).

- **(R5)** requires that the operation is intrinsic physicochemistry, not externally specified. DNA replication satisfies this: the polymerase's catalytic action is not driven by an external loss function. A neural network trained by gradient descent toward a target distribution does not satisfy (R5): the operation is externally specified by the loss function and is therefore outside the principle's scope. (R5) does not say the externally optimized system fails biology; it says the principle's structural analysis does not apply to it.

c. Theorem 1: necessity of the structural conditions. The theorem says: within the (R5) scope, any substrate-operation pair that supports open-ended information generation in the sense of Definition 1 must satisfy all four structural conditions (R1)–(R4). The claim is necessity, not sufficiency. Sufficiency would require additionally that the substrate's drift be navigable by selection on a connected fitness landscape, that the population be large enough to escape demographic stochasticity, and other downstream conditions outside the principle's scope.

Theorem 3 (Necessity of substrate recursion conditions)**.** *Within the scope defined by* (R5)*, if a substrate-operation pair* $(\mathcal{X}, \mathcal{O})$ *supports open-ended information-generating recursion in the sense of Definition 1, then it satisfies* (R1)*,* (R2)*,* (R3)*, and* (R4)*.*

The proof factorizes through two lemmas establishing the capacity bound. Let $C(L)$ denote the rate at which the configuration repertoire grows with L.

Lemma R1 (entropy per recognized site).: If (R1) holds, $h_{\text{site}} > 0$.

Lemma R2 (independently addressable non-recurrent sites).: If (R2) holds, $N(L) \to \infty$.

The capacity bound $C(L) \geq N(L) \cdot h_{\text{site}}$ requires both lemmas. Failure of (R1) gives $h_{\text{site}} = 0$ and the bound collapses to zero. Failure of (R2) gives bounded $N(L)$ and the capacity is bounded at $N_{\max} \cdot h_{\text{site}}$, which is constant in L. Failure of (R3) means $\mathcal{O}^2$ is not defined within $\mathcal{X}_L$, so the iteration terminates at the first application or transitions to a different substrate class on the second; the recursion is not closed within $(\mathcal{X}, \mathcal{O})$, and any heritability is parasitic on a different substrate that does satisfy (R3). Failure of (R4) has two sub-cases: zero drift gives a deterministic trajectory whose repertoire is at most countable in the iteration index; drift outside the recognition alphabet Σ produces variation that is not re-read by the next iteration's recognition step and therefore does not feed the recursion.

The four failures combine multiplicatively rather than additively. A substrate that satisfies (R1) and (R2) but fails (R3) cannot be rescued by its per-site entropy or its position count; the open-ended recursion requires the conjunction. Symbolically: open-endedness demands $C(L) \to \infty$ as $L \to \infty$ *and* the capacity to be reached through iterated $\mathcal{O}$-applications, with the iteration closed under (R3) and fed by drift through (R4). Each lemma or closure condition is necessary; together they are jointly necessary. The conjunction is what supports the $\Theta(2^{cL})$ scaling required by Definition 1; any single failure collapses the scaling to a strictly slower rate. The full proof, including the formal argument that the conjunction's scaling rate cannot be recovered from any subset of conditions, is in Supplementary Mathematical Appendix §5.

In biological terms: the theorem says that biology's open-endedness requires (a) the substrate to recognize multiple things at each site, (b) the site count to scale with the substrate's size parameter, (c) the operation's product to be readable by the same operation, and (d) the operation to produce drift in the alphabet it recognizes. Biology is not free to drop any of these and remain open-ended. Drt3b's bounded heredity is structural, not contingent: as long as the cyclic dynamics constrain the position count at the cycle length, the substrate cannot reach the open-ended regime regardless of how the population that hosts it evolves.

The principle is a substrate-level claim. The behaviors standardly described as natural selection, fitness, error thresholds, and evolutionary plateaus are downstream consequences. We state these as corollaries.

B. Corollary 1: bounded heredity under condition failure — the substrate-mode taxonomy

The necessity theorem says that any substrate failing one of (R1)–(R4) cannot reach the open-ended regime. It does not say such a substrate has no heredity. A substrate that fails one condition can still transmit information across iterations; what fails is the unboundedness of that transmission. The heredity is bounded, and the bound is determined by which condition fails. This is the content of Corollary 1, which immediately produces the substrate-mode taxonomy: each known biological templating mechanism corresponds to a specific failure pattern, and the bound on each mode's heredity is the quantitative consequence of that pattern.

Corollary 4 (Bounded heredity)**.** *If* $(\mathcal{X}, \mathcal{O})$ *is in the scope of (R5) and satisfies a strict subset of* (R1)–(R4)*, then the substrate may exhibit heredity, but the heredity is bounded in a manner determined by which conditions fail. Substrates failing (R1) have configuration-space ceilings determined by the per-site entropy (*$h_{\mathrm{site}} = 0$ *at the limit,* $h_{\mathrm{site}} \leq \log_2 K$ *for a* K*-state attractor); substrates failing (R2) have ceilings determined by bounded position count* $N_{\max}$ *(cycle length, module count); substrates failing (R3) inherit parasitically through an upstream substrate that satisfies (R3); substrates failing (R4) exhibit fixed configurations or non-recursive drift without information growth.*

The corollary's empirical instantiation is biology's substrate-mode inventory. We document six modes, each corresponding to a specific failure pattern of (R1)–(R4); information-capacity bounds and proofs appear in Supplementary Mathematical Appendix §2–3.

Mode 1: complementarity-based 1D sequence templating. Substrate: 1D linear sequence on a separate molecule. Mechanism: position-by-position chemical complementarity (Watson-Crick, β-sheet hydrogen bonding). Satisfies all four structural conditions (R1)–(R4) within (R5) scope: per-site entropy $h_{\mathrm{site}} = \log_2 |\mathcal{A}| > 0$, unbounded position count $N(L) = L$, daughter strand is reactant for next replication via complementarity (R3 satisfied via involution $\iota : X \to \bar{X}$), heritable local drift through the polymerase kernel. Capacity: $C_{\mathcal{S}}(L) = L \cdot \log_2 |\mathcal{A}|$ in the noise-free limit. Examples: DNA replication, Drt3a, retrons, telomerase, DRT9, β-sheet-mediated peptide replication.

Mode 2: code-encoded cross-descriptor templating. Substrate: 1D sequence in X plus separable lookup-table machinery in $\mathcal{O}$. Fails (R3) at the substrate level: protein products are not readable by the ribosome as templates for further translation. The recursion is recovered parasitically: the encoding gene for the operator (lookup table machinery) is itself a Mode 1 substrate. Capacity: $C_{\mathcal{S}}(L) \leq L \cdot H(\nu)$, where ν is the codon-count distribution over output symbols ($H(\nu) = 4.218$ bits/codon for the standard genetic code, including stop). Mode 2's heritability depends on Mode 1's recursion in the gene encoding the operator. Example: ribosomal translation.

Mode 3: cyclic-structure-encoded templating. Substrate: 3D protein structure with cyclic conformational dynamics having N distinct states. Fails (R2): position count is the cycle phase, bounded at N and recurrent ($N(L) = N$ regardless of L). The macro-level capacity is consequently bounded at $\log_2 N$. Output is periodic with period N. Examples: Drt3b producing $(\mathrm{AC})_n$ via $N = 2$ conformational states [5]. DRT7 represents the boundary case at $N = 1$, where $\log_2 1 = 0$: there is no positional information capacity, only structural channel selectivity. We classify DRT7 as a Mode 3 boundary / structural-selectivity limit rather than a positional-templating instance, since the $N = 1$ degenerate cycle is structurally a biased polymerase rather than a templating system.

Mode 4: conformer state-templating. Substrate: a discrete conformational state of a protein. Fails (R1) at the macro level: the recognition alphabet is the attractor repertoire of conformers, finite and typically small (K states). Fails (R2) trivially: the substrate has effectively a single "position" (the conformer assignment) rather than an extensible series. Capacity: $C_{\mathcal{S}}(L) \leq \log_2 K$, length-independent. Examples: prion propagation, amyloid self-conversion, DRT1 filamentous oligomer [6]. Prion strain repertoires confirm that some drift is transmitted across iterations, but the transmissible drift lies in a small attractor set rather than in an extensible alphabet.

Mode 5: modular conveyor templating. Substrate: spatial sequence of N structurally distinct modules within a single multi-modular assembly. Fails (R2) at the module count: position count is bounded by N, not extensible without adding modules to the assembly itself. Also fails (R3) at the substrate level: peptide products are not the assembly line; recursion is parasitic on Mode 1 acting on the gene cluster encoding the modules. Capacity: $C_{\mathcal{S}}(L) = N \cdot \log_2 k$ for $L \geq N$, with output length bounded by N. Examples: non-ribosomal peptide synthetases, polyketide synthases.

Mode 6: 2D surface-position templating. Substrate: spatial register on a 2D pattern. Mechanism: existing pattern biases the assembly of new units that themselves do not store the pattern. Mode 6 has extensible geometric positions (R2 is satisfied: surface position count grows with surface area) and per-site entropy in the unit-type alphabet (R1 is satisfied as long as multiple unit types are available). The failures are at (R3) and (R4): the 2D pattern does not autonomously self-copy with heritable local drift independent of genome-encoded parts. Pattern copying is not a same-operation recursion of the surface; it depends on Mode 1 acting on the genome to produce the constituent proteins, and pattern drift independent of the genome's drift cannot accumulate as heritable variation in the recursive sense. The biological instance is the eukaryotic ciliate cortex [3, 4]: cortical patterns of ciliary basal-body orientation are inherited at division through templating action of the existing cortex on newly assembled units; microsurgical experiments demonstrated that grafted cortical patches in inverted orientation propagate as inverted for hundreds of generations without genomic change. The framework retrodicts the cortex as a real but bounded inheritance system, parasitic on Mode 1 substrates that encode constituent proteins, in the same structural sense Mode 2 (translation) is parasitic on Mode 1.

The taxonomy is empirically anchored: every characterized DRT system fits one of Modes 1, 3, 4, plus the framework's out-of-scope boundary (§Tables). Modes 2, 5, 6 are anchored elsewhere (translation, NRPS/PKS, ciliate cortex respectively). The current inventory is provisional in the same sense any empirical taxonomy is: new biology may force additions, as Drt3b forced Mode 3 to be made explicit.

C. Corollary 2: population-dynamic plateaus from substrate condition failure

The first corollary classifies substrates by their structural failures. The second turns to the population dynamics those substrates host. The motivating question is whether the structural failures show up in the macroscopic behavior populations exhibit under selection. The principle predicts they do: a substrate that fails (R3) or (R4) at the individual level cannot accumulate the drift needed to climb a fitness landscape past a mechanism-specific ceiling, regardless of how the population is configured. This is a structural cap, not a limit imposed by population size, mutation load, or landscape topology — those would be conditions on top of the structural cap. We state this formally and then test it computationally.

Corollary 5 (Population-dynamic plateau)**.** *In finite populations evolving under selection toward a target phenotype on a tested fitness landscape, mechanisms whose substrate-operation pair fails (R3) or (R4) at the individual level are structurally capped at mechanism-specific ceilings* $F_k^* < F_{\max}$*, where the ceiling height is determined by the failed condition. Mechanisms satisfying (R3) and (R4) at the individual level are not structurally barred from cumulative climbing toward* $F_{\max}$*; whether climbing actually occurs depends on landscape topology, finite-population drift, mutation load, neutral-network connectivity, and other conditions outside the principle's scope.*

In biological terms: a population of perfect copiers (zero individual-level drift, R4 fails) cannot improve past the maximum fitness that was already present in its founding generation; selection sorts what exists but cannot generate new variants for selection to act on. A population of lineage-level copiers without individual-level drift (R3 holds at the lineage but the substrate operates only between lineage founders, not between individuals) faces a similar cap. Only populations whose substrate produces drift at the individual level can climb past the founding-generation ceiling. The corollary's positive direction is bounded: it says only that mechanisms satisfying (R3) and (R4) are not structurally barred from cumulative climbing; whether they actually climb depends on the landscape's connectivity and other conditions outside the principle.

We tested this in a pre-specified five-mechanism characterization[1] spanning the inheritance landscape from M0 (no recursion: fails R3 and R4) to M4 (canonical Mode 1: satisfies R1–R4 within R5 scope). M0: pure stateless — phenotype freshly drawn each generation, no lineage state. M1: lineage-level fixation — at lineage founding, draw a random template; lineage members all share that template; reproduction copies the lineage label, no individual-level drift (satisfies R3 at lineage level, fails R4). M2: lineage-level fixation with re-draw rate $r \in [0, 1]$ (satisfies R3 at lineage level, partial R4 via wholesale re-draw rather than local drift). M3: individual-level exact copy without drift (satisfies R3 at individual level, fails R4). M4: individual-level copy with local drift (satisfies R3 and R4 at individual level).

Plateau heights at gen 1000 averaged across 30 replicates, with target length $L = 32$, $K = 400$ agents, $\beta = 10$ selection sharpness: M0 = 0.249 (chance), M1 = 0.472, M3 = 0.498, M2 at $r = 0.10 = 0.533$, M4 = 0.972 (Figure 5A; Table V). Only M4 reaches near-target fitness in the tested model class. M1 and M3 plateau at the maximum fitness present in the initial-population's lineage repertoire — the (R4) failure produces no new drift. M2 has a regime-invariant sweet spot at $r = 0.10$: too little re-draw and the system cannot escape sub-optimal initial draws; too much and the lineage advantage is washed out. The sweet spot's location is invariant within 0.05 across $\beta \in \{2, 5, 10, 20\}$ and $L \in \{16, 32, 64, 128\}$; it is a real local optimum, not noise.

A direct head-to-head test confirms the corollary's content: lineage-level re-draws and individual-level local drift are not interchangeable input sources in the tested model. M2 at $r^* = 0.10$ introduces 2.40 expected new positions per offspring; M4 at $\mu^* = 0.001$ (its tested-optimal kernel rate) introduces 0.02 new positions per offspring. M2's drift rate is 100× M4's, yet M2 reaches plateau 0.548 while M4 reaches 0.997. In direct competition at balanced start, M4 wins 67%, M2 wins 33%, no ties. *Targeted local drift (R4 at the individual level) converts input into fitness gain more efficiently than wholesale lineage re-draw in the tested landscape.* The structural distinction matters; rate alone does not.

The qualitative claim survives at long horizons in the tested model. At extended simulation (5000 generations, equal-start with $K_{M1} = K_{M4} = 200$), M4 wins outright in 24/30 replicates against M1, with the remaining 6/30 reflecting cases where M1 fixes early before M4's drift-and-selection cycle produces a winning lineage. M1's plateau is durable but bounded; the limited-heredity ceiling is a structural cap, not a transient state. At more

[1] Analysis plans for each test cohort were written in dispatch markdown files (`dispatch1.md` through `dispatches_v4/`) and reproduced verbatim in the docstring header of each test script before the corresponding result CSVs were generated. Test script docstrings (e.g. `test_h_competition.py` lines 18–38, `test_h3_inheritance_landscape.py`, `test_h4_long_horizon.py`) name each prediction (P_H1–P_H4 etc.) with its falsification criterion, and per-test `*_predictions_v1.md` files compare each prediction to its empirical result. The full deposit is archived at 10.5281/zenodo.20060972 (Phase 0 retroactively committing the dispatches; Phase 1–5 commits adding scripts and result CSVs).

asymmetric initial conditions ($K_{M1} = 80$, $K_{M4} = 320$ or $K_{M1} = 20$, $K_{M4} = 380$), M4 wins 30/30, confirming that the structural cap on M1 holds robustly when M4 is given the population-size conditions to actually exploit its (R3, R4) satisfaction.

The corollary's prediction for the eukaryotic ciliate cortex is consistent with empirical observation (Figure 6). The cortex is built each generation from genome-encoded proteins (re-draw at every cell division) but the existing pattern biases the assembly (lineage-level fixation). This is operationally close to M2 with low re-draw rate — exactly the regime where the framework predicts limited but heritable variation. Cortical inheritance has been documented to propagate non-genomic structural variation for hundreds of generations [3], consistent with M2's plateau persistence; cortical lineages have not been observed to accumulate qualitatively new structures over long timescales, consistent with M2's bounded ceiling.

D. A descriptor-relative apparatus distinguishes templating from biasing-participants

The principle's empirical anchoring requires an instrument that classifies templating events by mode. We define a templating event as the tuple $(X, \mathcal{O}, S, \Delta G; \varphi_X, \varphi_Y) \to Y$, where X is the template, $\mathcal{O}$ is the operator, S is the substrate pool, ΔG is the free-energy driving, and φ_X, φ_Y are descriptor maps.

The diagnostic apparatus is a joint observable in three coordinates. The first is *population mutual information*,

$$I^{\mathrm{pop}}_{\mathrm{struct}}(X;Y) = H(\varphi_Y(Y)) - \mathbb{E}_{X\sim\Pi}\left[H(\varphi_Y(Y) \mid X)\right], \tag{1}$$

where the expectation is taken over a stated template population $\mathcal{X}$ with distribution Π. $I^{\mathrm{pop}}_{\mathrm{struct}}$ is bounded above by $H(\Pi)$ and is identically zero when $\mathcal{X}$ is degenerate. This measures *realized* information transfer in a stated population, distinct from substrate capacity (defined below).

The second is *channel mutual information* against a stated comparison ensemble $\mathcal{C}$ of templating channels,

$$I^{\mathrm{chan}}_{\mathrm{struct}}(\mathcal{C}) = \sum_{c\in\mathcal{C}} P(C=c)\, D_{\mathrm{KL}}\big(P(\varphi_Y(Y) \mid C=c) \,\|\, \bar{P}(\varphi_Y(Y))\big), \tag{2}$$

where $\bar{P}(\varphi_Y(Y)) = \sum_c P(C=c)P(\varphi_Y(Y) \mid C=c)$ is the mixture distribution. Unlike $I^{\mathrm{pop}}_{\mathrm{struct}}$, $I^{\mathrm{chan}}_{\mathrm{struct}}$ does not vanish for fixed-template channels: a channel with a single fixed template still produces a distinguishable output distribution against alternative channels.

The third is *per-base mechanism fidelity*, resolved into two distinct components when the channel has cyclic structure. *Phase fidelity* f_{phase} is the fidelity of cycle-state alternation; *monomer fidelity* f_{monomer} is the fidelity of correct monomer choice within each state. The triple $(I^{\mathrm{pop}}_{\mathrm{struct}}, I^{\mathrm{chan}}_{\mathrm{struct}}, (f_{\mathrm{phase}}, f_{\mathrm{monomer}}))$ is the apparatus signature; mode classification uses the joint signature.

We distinguish realized population MI $I^{\mathrm{pop}}_{\mathrm{struct}}$ from *substrate capacity*

$$C_{\mathcal{S}}(L) = \sup_{\Pi_X} I^{\mathrm{pop}}_{\mathrm{struct}}(X;Y), \tag{3}$$

a property of the substrate class $\mathcal{S}$ rather than of any specific population. $C_{\mathcal{S}}$ is what enters the principle's (R1) condition; $I^{\mathrm{pop}}_{\mathrm{struct}}$ is what the apparatus measures on a particular biological system. Both are needed; conflating them is a common failure mode of substrate-level reasoning.

A claim that X is a structurally specific template requires two controls. The bulk-matched control X_{bulk} produces an output Y_{bulk} with kth-order matched marginals but no position-specific information. The structure-scrambled control Y_{scram} has the same composition as Y with positional pattern shuffled. Real templating must give $I^{\mathrm{pop}}_{\mathrm{struct}}(X;Y) \gg I^{\mathrm{pop}}_{\mathrm{struct}}(X;Y_{\mathrm{bulk}})$ and $I^{\mathrm{pop}}_{\mathrm{struct}}(X;Y) \gg I^{\mathrm{pop}}_{\mathrm{struct}}(X;Y_{\mathrm{scram}})$ to a stated significance threshold.

We validate the apparatus on the canonical case. A simulated Watson-Crick replication system with template length L, alphabet size 4, and per-position misincorporation rate ε should give $I^{\mathrm{pop}}_{\mathrm{struct}} = L \cdot [2 - h_b(\varepsilon) - \varepsilon \log_2 3]$. Across 30 sweep cells, the empirical $I^{\mathrm{pop}}_{\mathrm{struct}}$ matches theory to within 0.6% (Figure 1A). The bulk-matched control discriminates cleanly: across 24 non-uniform sweep cells, separation ratios between genuine templating and bulk-matched control range 450× to 1500× (Figure 1B). The apparatus signature is robust to the choice of comparison ensemble for $I^{\mathrm{chan}}_{\mathrm{struct}}$: across 10 random four-channel ensembles drawn from a 10-channel pool, every channel's mode classification was stable.

E. Mode 3 saturates at $\log_2 N$; Mode 5 is bounded by module count; Mode 1 scales linearly

We tested the corollary's mode-specific capacity predictions directly. A two-state cyclic active site, sweeping $N \in \{2, 3, 4, 5, 6, 8, 10\}$ with low-noise channel ($\varepsilon = 10^{-3}$), should approach $I^{\mathrm{pop}}_{\mathrm{struct}} = \log_2 N$ from below. Empirical measurements approach the bound to within estimator precision: at $N = 2$, $I^{\mathrm{pop}}_{\mathrm{struct}} = 1.0000$ bits; at $N = 5$, 2.3219 bits; at $N = 10$, 3.3219 bits (Figure 2A). For each N, $I^{\mathrm{pop}}_{\mathrm{struct}}$ is constant across L from $L = N$ to $L = 50N$, with maximum drift below 0.0001%. Mode 3 information saturates while Mode 1 information scales linearly with L (Figure 2B), confirming the framework's extensibility-failure prediction within finite-sample noise.

The autocorrelation of Mode 3 output peaks at integer multiples of N, with peak value matching the noise-model prediction $(1-\varepsilon)^2 + \varepsilon^2/(N-1)$ (Figure 2C). Mode 5 simulation, sweeping module count $N \in \{2, 4, 8, 16, 32\}$ at alphabet sizes 4 and 20, confirms linear scaling in N. The length limit is sharp: positions beyond the module count carry zero information about the template (Figure 2D).

For Mode 2, we simulated the standard genetic code applied to uniformly random DNA templates of length L codons. The empirical $I^{\text{pop}}_{\text{struct}}$ matches the analytical bound $L \cdot H(\nu) = 4.218L$ bits within finite-sample noise (4.214 to four significant figures at $L = 64$, $n = 1000$). The Mode 1-vs-Mode 2 capacity ratio at matched DNA length is $6/4.218 \approx 1.42$. A nested two-level evolution simulation — selection on phenotype, with the codon-to-amino-acid lookup table itself encoded as a Mode 1 substrate — produced the predicted dynamic: 500 generations drove population fitness from 5% to 43%, with code fidelity converging to 52% as selection acted on phenotype while the recursion acted on the gene encoding the code. The simulation computationally illustrates the mechanism by which Mode 2 inherits parasitically through Mode 1 acting on the encoding gene; it does not by itself confirm the biological claim about translation, which rests on independently established molecular biology.

F. The framework anchors quantitatively to the Drt3 system

The Drt3 system [5] provides a direct biological test. The complex contains two reverse transcriptases and a noncoding RNA. Drt3a uses a conserved ACACAC region of the ncRNA as Watson-Crick template (Mode 1) to produce poly(GT) DNA. Drt3b synthesizes the complementary poly(AC) strand without any nucleic acid template, using two amino-acid residues — Glu26 (gatekeeper for dA selection through hydrogen-bonding to the N6 amine of dA) and Arg253 (cation-π stabilizer for dA, plus three hydrogen bonds with the Watson-Crick edge of dC) — to enforce alternation. The E26Q point mutation perturbs dA gating selectively, and the in vitro mutant produces alternating poly(AC) with approximately 80% dA / 20% dG misincorporation at the dA-selecting position while preserving dC selectivity [5, Fig. 4J]. The same protein fold appears in AbiK with different active-site residues, where the system produces random ssDNA [5].

We classify these systems by parameterizing Markov-chain channels from the published biochemistry and asking whether the apparatus recovers the expected mode classifications.

Drt3b WT, parameterized from the (AC) alternating product, gives $I^{\text{chan}}_{\text{struct}} = 1.0000$ bits against the standard 5-channel ensemble. Phase fidelity $f_{\text{phase}} \approx 0.99$ (cycle alternation intact) and monomer fidelity $f_{\text{monomer}} \approx 0.99$ at both states. Periodicity peaks at lag 2 with peak autocorrelation 0.9802. Output marginal frequencies: 0.497 dA, 0.497 dC, 0.003 dG, 0.003 dT. Separation ratio against bulk-matched control ranges $234\times$ to $1031\times$.

Drt3b E26Q, parameterized with the paper's 80%/20% dA/dG split at the dA-selecting state, gives $I^{\text{chan}}_{\text{struct}} = 1.0000$ bits and $f_{\text{phase}} \approx 0.99$ (cycle architecture preserved) but $f_{\text{monomer}} \approx 0.80$ at the A-state (monomer selectivity at A-state degraded; C-state unchanged). Marginal G fraction rises to 0.102, exactly matching the analytical prediction $0.5 \times 0.20 = 0.10$ for half-cycle dG misincorporation; periodicity peak drops to 0.83; separation ratio against bulk-matched control falls to $\sim 100\times$ (Figure 3A,B). The apparatus classifies E26Q as Mode 3 with degraded monomer selectivity at one state but intact phase architecture — a distinction motivating the architectural-vs-selectivity gate classification below.

AbiK same-fold non-templating, parameterized from the published random-product description, gives $I^{\text{pop}}_{\text{struct}} \approx 0.03$ bits, $I^{\text{chan}}_{\text{struct}} < 0.01$ bits, $f_{\text{monomer}} \approx 0.25$ (random over four nucleotides). The apparatus correctly excludes AbiK from Mode 3 and from any structurally specific template class.

Drt3a Mode 1 versus Drt3b Mode 3, treated as distinct channels in a 5-channel comparison ensemble, separate by $\Delta I^{\text{chan}}_{\text{struct}} = 1.0$ bits against the stated five-channel ensemble despite producing nearly identical alternating output (Figure 3C). The channel-as-X observable distinguishes mechanism even when output statistics match: Drt3a's f_{monomer} characterizes Watson-Crick fidelity against an external ACACAC template; Drt3b's f_{phase} and f_{monomer} characterize the cyclic active-site channel. The numerical separation is ensemble-dependent — a different comparison ensemble would yield a different number — but the joint signature's qualitative discrimination is stable across ensemble choices, as shown by the robustness check above. The apparatus's joint signature, not any scalar summary, is what enables the discrimination.

The methodological implication is that the diagnostic apparatus is intrinsically multi-channel. WT Drt3b and E26Q both saturate $I^{\text{chan}}_{\text{struct}}$ at 1 bit but differ sharply in marginal G fraction (0.003 versus 0.102) and periodicity peak (0.98 versus 0.83). E26Q and AbiK both produce non-WT compositions but differ in $I^{\text{pop}}_{\text{struct}}$ (1.0 versus 0.03) and f_{monomer} (0.99 versus 0.25). Drt3a and Drt3b separate at the channel-distinctness level. Collapsing the joint signature to a single score discards the information that distinguishes the systems.

G. Drt3b family-level apparatus application

We applied the apparatus across 1,232 Drt3b homologs identified from the Deng et al. supplementary alignment. Each homolog was assigned to a clade by primary-gate residue (E26, Q26, D26, other) and secondary-residue identity at six conserved positions identified through structural alignment. The framework's residue partition follows the Deng paper's structural assignments: R253 (universally invariant) is the architectural gate enforcing cycle phase via cation-π stabilization of the templating dC; G248 (essentially invariant) is a selectivity gate at the C-state via steric exclusion of dG; E26 is a primary selectivity gate at the A-state, mimicking a templating nucleobase via two hydrogen bonds with the N^6 amine of incoming dA [5]. Family-level apparatus application predicts: clades with WT-like primary and secondary residues should fall in-envelope at the apparatus signature observed for WT Drt3b ($I^{\text{chan}}_{\text{struct}} = 1.0$ bits separation against the stated five-channel ensemble, marginal G fraction < 0.005). Clades with E26→D primary-gate substitution shorten the carboxylate side chain and degrade the selectivity of the A-state channel; the framework predicts elevated dG misincorporation at the A-state position, with the cycle architecture preserved (R253 is invariant in all six E26→D clades).

Of 33 secondary-residue clades assayed, all 33 fall in-envelope (Figure 4A). Of 6 E26→D primary-gate clades (C06, C14, C20, C27, C28, C42), all 6 show the framework-predicted shift: marginal G fraction rises from ~ 0.003 in WT-like clades to ~ 0.052 across the six E26→D clades — a $\sim$ 17-fold elevation in dG misincorporation at the A-state, consistent with the residual hydrogen-bonding flexibility allowed by the shortened Asp side chain (Figure 4B). The periodicity peak drops from ~ 0.98 in WT-like clades to ~ 0.857 in the E26→D clades; this drop is the arithmetic consequence of A-state selectivity loss, not an independent architectural failure. With state-A residency probabilities $(0.85, 0.10, 0.025, 0.025)$ over (A, G, C, T) and state-C unchanged at 0.99 for C, the analytical period-2 self-match probability is $\frac{1}{2}\sum_i p_A(i)^2 + \frac{1}{2}\sum_i p_C(i)^2 \approx 0.857$, matching the simulation to three decimal places. The cycle architecture itself is intact in all six E26→D clades (R253 invariant), and the apparatus correctly localizes the failure to the A-state selectivity channel rather than to the cycle's phase structure. The family-level apparatus application confirms the Mode 3 classification holds across the Drt3b family rather than only at the Drt3b reference sequence, and the selectivity-only signature distinguishes E26 from R253, the universal architectural gate whose substitution is predicted to collapse the cycle entirely.

H. Universal-gate site-directed-mutagenesis predictions

The architectural-vs-selectivity distinction at the apparatus level translates into structurally distinct site-directed mutagenesis predictions, with the family-level E26→D data providing an empirical anchor on the selectivity side. R253 is an architectural gate: its substitution disrupts the cation-π stabilization that holds the cycle in its alternating phase, predicted to collapse $I^{\text{pop}}_{\text{struct}}$ to near-zero with periodicity peak dropping to ~ 0.43. G248 is a selectivity gate at the C-state: its substitution alters monomer selectivity within an intact cycle, predicted to preserve $I^{\text{pop}}_{\text{struct}}$ near WT while elevating marginal G to ~ 0.15. E26 is a selectivity gate at the A-state, structurally analogous to G248 but on the complementary half-cycle; the family-level E26→D data confirm the selectivity-gate signature empirically (marginal G ~ 0.052, cycle phase preserved). The R253A and G248A SDM predictions therefore have an empirical anchor for the selectivity side already; R253A is the cleanest single-experiment falsification target the framework offers, with 100× predicted separation from WT $I^{\text{pop}}_{\text{struct}}$.

R253A is predicted to collapse $I^{\text{pop}}_{\text{struct}}$ to 0.0090 bits (from WT 0.9997 bits), with marginal G fraction rising to 0.1265 and periodicity peak dropping to 0.4325. The cycle architecture is destroyed; the system is reduced to compositional bias only. If observed $I^{\text{pop}}_{\text{struct}} > 0.5$ bits, the R253-as-architectural-gate claim is wrong.

G248A is predicted to preserve $I^{\text{pop}}_{\text{struct}}$ at 0.9966 bits, with marginal G fraction rising to 0.1516 and periodicity peak dropping to 0.7469. The cycle is intact; the dG misincorporation comes from a selectivity shift at one state, not from architectural collapse. If observed $I^{\text{pop}}_{\text{struct}} < 0.5$ bits, G248 is architectural and the gate-class distinction fails.

Predictions for partial substitutions (R253K, R253H, G248V, G248D) and the double mutant (R253A + G248A) are stated in Table IV. The double mutant is the strongest combined falsification target: if G248A signature dominates over R253A's architectural collapse, the gate-class ordering is wrong; under the framework's prediction, R253A's architectural collapse should dominate the joint signature.

III. DISCUSSION

The framework classifies biology's known templating mechanisms as a small structured inventory. DNA replication, telomerase, retron reverse transcription, Drt3a, and the β-sheet peptide replicators [1] sit together as Mode 1: the canonical 1D sequence-template substrate satisfying all four structural conditions within the (R5) scope. Ribosomal translation sits as Mode 2: a substrate that fails (R3) at the protein-product level and recovers recursion only parasitically through Mode 1 acting on the gene that encodes the ribosomal machinery. Drt3b sits as Mode 3: a cyclic-conformational substrate whose two-phase active-site dynamics cap its information capacity at $\log_2 2 = 1$ bit regardless of product length. DRT7 sits at the Mode 3 boundary at $N = 1$ where positional capacity vanishes entirely. Prions and amyloid-templated peptide synthesis sit as Mode 4: a small attractor repertoire transmitting drift across iterations within a bounded conformational alphabet. NRPS and polyketide synthases sit as Mode 5: a modular conveyor whose position count is bounded at the module count and whose recursion is again parasitic on Mode 1. The eukaryotic ciliate cortex sits as Mode 6: a 2D surface-position substrate that satisfies (R1) and (R2) at the surface level but fails (R3) and (R4) because the surface does not autonomously self-copy and pattern drift independent of genome drift cannot accumulate as heritable variation. The mechanisms biology has historically described in mutually independent vocabularies turn out to share a small set of structural failure patterns; what differs is which pattern each mechanism instantiates.

The substrate recursion principle gives biology a structural specification it has lacked. The four structural conditions (R1)–(R4) are properties of the substrate-operation pair within the (R5) scope, not of the population-level dynamics that the substrate hosts. Population behaviors that the literature has historically taken as primary — selection, fitness landscapes, error catastrophe, evolutionary plateaus, lineage formation — are downstream consequences of substrates that satisfy or fail the structural conditions. Re-grounding the analysis at the substrate level reorganizes the field's conceptual structure: the question "what supports biology?" becomes "what conditions must a physical substrate-operation pair satisfy to host biology's recursion?" The answer, in four structural conditions plus a scope-defining condition, is more constrained than the literature has acknowledged.

The principle commits to a substrate-essentialist position with one further entailment that the manuscript has stated explicitly: the recursion has no privileged starting configuration. Every template is the drifted product of prior recursions, and the substrate-level analysis treats no configuration as canonical. This commitment shows up vocabularily as the use of "drift" rather than "error" or "variant" wherever the latter would smuggle a target back in. "Error" presupposes a correct answer the operation was trying for; in a non-teleological substrate, there is no correct answer, only what the physics produces. The biology of replication is the same: the polymerase produces a distribution over outcomes, and the consensus sequence is a moving average over a population's drifts, not a Platonic ideal. The terminological move is consistent with the (R5) commitment in a way "error" is not. We retain "error" for the field-standard usage of Eigen's error threshold and the per-position misincorporation rate ε against a defined Watson-Crick canonical product, where the technical meaning is established and the term cannot be replaced without losing precision.

The principle's substrate-essentialism is balanced by descriptor-relativity at the apparatus level. The conditions (R1)–(R4) are stated structurally, but operationalizing them requires a descriptor choice (φ_X, φ_Y). The apparatus is descriptor-relative; the classification is conditional on the descriptor declared up-front. This is not a weakness; it is the appropriate level of operational constraint. A descriptor-free taxonomy would be either trivially true (all biology is templated) or impossible to test (no apparatus can be defined). The framework's commitment is that the conditions, given an admissible descriptor choice, generate stable mode classifications across reasonable descriptor variations. We have demonstrated this stability for the Drt3b case: descriptor choices spanning the natural range produce the same Mode 3 classification.

The principle's relationship to natural selection is one of layered explanation. Selection is real as a description of which substrate-instances persist in finite populations under environmental filtering. Selection is not, in this framework, a force in the substrate. The substrate runs because its physicochemistry runs; certain configurations carry forward because they are produced by the operation; certain configurations fail to carry forward because they are not produced or are produced unstably. Selection-talk is descriptive vocabulary applied to population-level outcomes of substrate-level recursion. This is a deflationary reading of selection's causal role at the substrate level, not a denial of selection's empirical reality.

The principle has consequences for the closure tradition. Rosen [18], Maturana and Varela [19], Pattee [20], and Hofmeyr [21] ask how a system produces its own templates, the operators that read those templates, and the metabolic cycles that produce both. The principle is upstream of these questions: it specifies what physical substrates can serve as templates, given which closure becomes a tractable question. The two are complementary. Closure-capable systems must use inheritance-capable substrates; the principle's Modes 1 and 2 are the only options.

The strongest empirical challenge in the prior literature is yeast prion biology [22, 23]. Prion conformations are real, ecologically prevalent, and evolutionarily active in wild yeast populations: they generate heritable phenotypic variation that selection acts on. The framework's response is not that prion inheritance does not exist, but that prion inheritance is parasitic on Mode 1: the variation transmitted is conformational state of a polypeptide whose primary structure is genome-encoded; the prion can stabilize or accelerate adaptation by exposing cryptic genetic variation, but the substrate of inheritance — the gene encoding the prion-prone protein — is sequence. Prions function as transient capacitors of evolutionary potential, not as independent inheritance carriers. The framework's classification is consistent with this empirical literature.

The strongest formal challenge is collectively autocatalytic sets of the GARD/compositional-inheritance class [14, 15]. Vasas *et al.* [17] showed using the GARD model itself that compositional inheritance fails the fidelity requirements for open-ended evolution: high-fidelity compositional transmission requires a lognormal distribution of catalytic rate constants so narrow as to be biologically unrealistic. The principle's analysis aligns with this verdict: GARD-class compositional inheritance fails (R2) (no addressable position structure, only composition vectors lacking $N(L) \to \infty$) and (R4) (drift in composition is not drift in the next iteration's recognized alphabet, because compositional inheritance has no per-site recognition step). Vasas *et al.* [17] from within the metabolism-first camp concluded that "this fundamental limitation of ensemble replicators cautions against metabolism-first theories of the origin of life." The principle gives the structural reason behind the empirical failure.

The Mode 6 reframing has been substantively revised from earlier framings of this work. The view that 2D surface-position templating is absent from biology was empirically wrong: the eukaryotic ciliate cortex has been documented since 1965 as a non-genomic structural inheritance system. The corrected claim, supported by the principle, is that Mode 6 exists in biology as a limited-heredity supplementary system, dependent on Mode 1 substrates that encode constituent proteins. The cortex satisfies (R1) and (R2) at the surface level but fails (R3) and (R4): the surface does not autonomously self-copy with heritable local drift independent of genome-encoded parts; pattern copying depends on Mode 1 acting on the genome to produce the constituent proteins, and pattern drift independent of genome drift cannot accumulate as heritable variation in the recursive sense. The cortex is parasitic on Mode 1 in the same structural sense Mode 2 (translation) is parasitic on Mode 1: the operator that maintains the cortical pattern is itself a Mode 1 substrate's product. The cortex is not a counterexample; it is a positive prediction the principle retrodicts.

The framework has limitations stated as limitations. The biological inventory of six modes is provisional; new biology may force additions. The taxonomy is descriptive in the sense that the modes are derived from attested examples rather than from a deductive enumeration of all possible substrates that satisfy or partially satisfy (R1)–(R4); a more principled taxonomy would derive modes from substrate dimensionality, recognition cardinality, and recursion topology as orthogonal axes within the principle. The current framework presents the modes as attested categories, with the joint observable as the operational classifier. The predictions have been tested in coarse-grained Markov-chain simulation; an atomistic model of Drt3b's active-site cycle would test whether the cyclic-state picture survives realistic energy landscapes. The classification of the DRT family rests on published mechanism descriptions rather than direct apparatus application to raw sequencing data; the next step requires access to deposited reads which the Deng paper does not provide for the DRT3 product specifically. (R5) is scope-defining rather than structural: it demarcates the domain in which (R1)–(R4) apply rather than being itself a structural condition with an information-theoretic bound. Systems whose operation is externally optimized (gradient descent, designed catalytic cycles) are outside the principle's scope rather than instances of structural failure within it. This is the appropriate formal status; treating (R5) as a fifth structural condition would conflate domain restriction with structural necessity.

The framework's value depends on whether future templating-system discoveries can be systematically classified within its inventory and whether the principle's conditions hold up across novel substrates. Our family analysis of 1,232 Drt3b homologs, our placement of seven other DRT systems, our Mode 6 reframing, and our population-dynamics characterization of bounded-heredity regimes make this an empirical bet rather than a stipulation. The architectural-vs-selectivity gate distinction generates two single-experiment falsification targets in R253A and G248A SDM. The principle's mechanism-specific theorem aligns with the metabolism-first camp's own refutation of compositional autocatalysis. If new discoveries continue to fit, the framework provides predictive scaffolding. If they keep requiring new modes, the framework's classificatory power is weaker than we have argued. The bet is well-calibrated by the evidence presented here, and natural history will verify it.

IV. MATERIALS AND METHODS

A. Mathematical setup

A templating event is the tuple $(X, \mathcal{O}, S, \Delta G; \varphi_X, \varphi_Y) \rightarrow Y$ where X is the template, $\mathcal{O}$ is the operator (separable molecular machinery, possibly empty or coincident with X), S is the substrate pool, ΔG is the free-energy budget, and φ_X, φ_Y are descriptor maps on template and product spaces. When $\varphi_X = \varphi_Y$ the templating is same-descriptor; otherwise it is cross-descriptor and $\mathcal{O}$ implements the coupling.

The substrate recursion principle's conditions (R1)–(R5) are formalized in Supplementary Mathematical Appendix §5. The diagnostic apparatus is a triple $(I^{\text{pop}}_{\text{struct}}, I^{\text{chan}}_{\text{struct}}, (f_{\text{phase}}, f_{\text{monomer}}))$ defined formally in Supplementary Mathematical Appendix §1. We summarize the operational definitions used in the main paper. Population mutual information $I^{\text{pop}}_{\text{struct}} = H(\varphi_Y(Y)) - \mathbb{E}_X[H(\varphi_Y(Y) \mid X)]$ is estimated via plug-in entropy estimators on empirical joint distributions; channel mutual information $I^{\text{chan}}_{\text{struct}}(\mathcal{C})$ is estimated as the mixture-MI under a stated comparison ensemble using $n_{\text{samples}} \geq 5000$ per channel. Per-base fidelities f_{phase} (cycle-state alternation) and f_{monomer} (canonical-monomer selection within a state) are estimated as empirical correctness rates against canonical-output functions. The substrate capacity $C_{\mathcal{S}}(L) = \sup_{\Pi_X} I(X; X')$ is computed analytically per mode (Appendix §2) rather than estimated.

The bulk-matched control $X_{\text{bulk}}, Y_{\text{bulk}}$ preserves kth-order statistics of Y. For $k = 1$, Y_{bulk} is drawn iid from the empirical first-order composition. For $k = 2$, Y_{bulk} is drawn from the maximum-entropy first-order Markov chain preserving empirical nearest-neighbor frequencies. The structure-scrambled control Y_{scram} has the same composition as Y with positional pattern shuffled.

B. Simulation framework

Simulations were implemented in Python 3.10 using NumPy 1.24 and Matplotlib 3.7. Random seeds were set explicitly for reproducibility (per-replicate seed = 42 + cell_idx × 100 + rep_idx). Computations ran on a 64-CPU compute node. Code and raw simulation outputs are available at 10.5281/zenodo.20060972.

C. Test A.1 — Mode 1 length-scaling validation

Templates X of length L over alphabet $\{A, C, G, T\}$ drawn uniformly. Products Y generated by Watson-Crick channel with per-position misincorporation rate ε. Sweep $\varepsilon \in \{0.001, 0.01, 0.05, 0.10, 0.25\}$ and $L \in \{10, 25, 50, 100, 200, 500\}$ with $n = 5000$ per cell. Per-position MI $I(X_i; Y_i)$ computed by plug-in estimator. Theoretical comparison: $I_{\text{struct}}^{\text{pop}} = L \cdot [2 - h_b(\varepsilon) - \varepsilon \log_2 3]$.

D. Test A.2 — bulk-matched control discrimination

Three template biases tested (π_{uniform}, $\pi_{\text{AT-skew}}$, $\pi_{\text{GC-skew}}$). Bulk control draws Y_{bulk} iid from analytical output marginal Q at each position, with X_{bulk} uncorrelated. Pass: $I_{\text{struct}}^{\text{pop}}{}^{\text{bulk}} < 0.05$ bits, separation ratio $> 20\times$.

E. Test B — Mode 3 information saturation

Two-state cyclic active site simulation. Sweep $N \in \{2, 3, 4, 5, 6, 8, 10\}$, $L \in \{N, 5N, 10N, 50N\}$, $\varepsilon = 10^{-3}$, $n = 5000$ per cell. $I_{\text{struct}}^{\text{pop}}$ computed against template ensemble = phase index. Theoretical bound: $I_{\text{struct}}^{\text{pop}} \leq \log_2 N$.

F. Test C — Mode 5 module-bounded scaling

Modular conveyor channels with N modules, alphabet sizes $|\mathcal{A}| \in \{4, 20\}$. Sweep $N \in \{2, 4, 8, 16, 32\}$, $L = 5N$, $n = 5000$ per cell. Theoretical bound: $I_{\text{struct}}^{\text{pop}} = N \cdot \log_2 |\mathcal{A}|$ for $L \geq N$.

G. Test D — Mode 2 codon-degeneracy capacity

Standard genetic code (NCBI table 1). Random DNA templates length L codons, $L \in \{16, 32, 64, 128\}$, $n = 1000$ per cell. Output measured per amino acid (with stop). Theoretical bound: $L \cdot H(\nu) = 4.218L$ bits.

H. Test F — apparatus signature on Drt3 channels

Five-channel ensemble: {Drt3a Mode 1, Drt3b Mode 3 WT, Drt3b E26Q, Drt3a homopolymer null, AbiK random}. Channel ensemble robustness: 10 alternative four-channel ensembles drawn from a 10-channel pool. Mode classification stability checked across all ensemble draws.

I. Test G — Drt3b homolog family-level apparatus application

1,232 Drt3b homologs from Deng et al. supplementary alignment. Clade assignment by primary-gate residue (E26, Q26, D26, other) and secondary-residue identity at six conserved positions. Apparatus signature predicted per clade; in-envelope assessment at the WT-Drt3b reference signature.

J. Test H — five-mechanism population-dynamics characterization

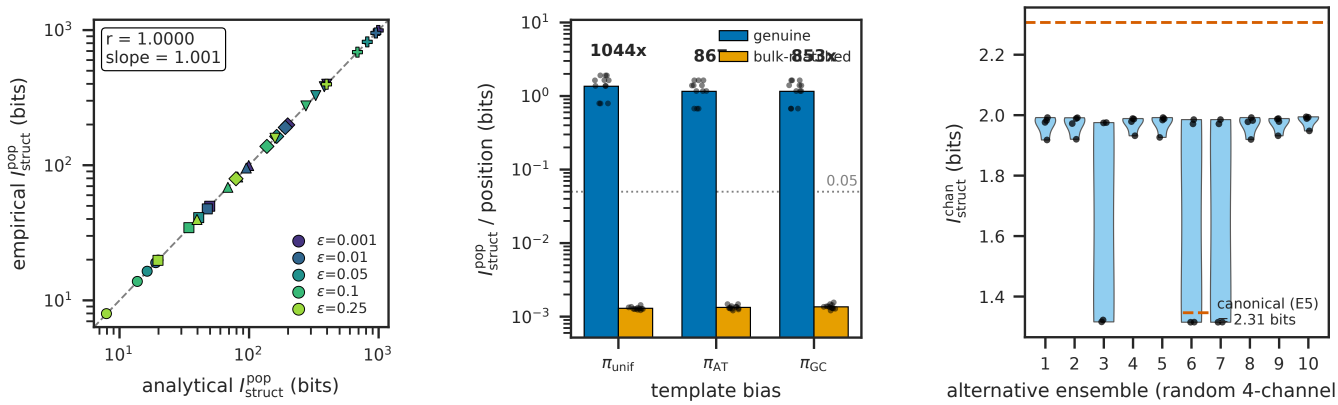
Mechanisms M0–M4 as defined in Results. Target length $L_{\text{TARGET}} \in \{16, 32, 64, 128\}$. Population size $K = 400$. Selection sharpness $\beta \in \{2, 5, 10, 20\}$. Mutation rate (M4) $\mu \in \{0.001, 0.01, 0.05, 0.10\}$. Re-draw rate (M2) $r \in \{0.01, 0.05, 0.10, 0.25, 0.50\}$. 30 replicates per cell. Plateau height = mean fitness at gen 1000. Direct competition (Test H4) uses balanced start, equal initial population sizes; head-to-head winner = 5000-gen majority. Long-horizon stability (Test H5) at 5000 generations.

K. Test I — universal-gate SDM predictions

R253 substitutions: R253A, R253K, R253H. G248 substitutions: G248A, G248V, G248D. Double mutant: R253A + G248A. Markov-chain channels parameterized from structural homology arguments and Drt3b WT reference. $L = 64$, 30 replicates, $n_{\text{samples}} = 5000$ per replicate. 95% CIs from beta-binomial.

DATA AVAILABILITY

All simulation code, raw simulation outputs, the diagnostic apparatus implementation, the family-level analysis pipeline, and the dispatch markdown files containing the pre-specified analysis plans for each test cohort are deposited in a permanent public repository archived at 10.5281/zenodo.20060972. Phylogenetic input files (`science_aed1656_data_s4--s8`) are derived from the published supplementary material of Deng *et al.* [5]. The deposit's git history records the commit ordering: Phase 0 retroactively commits the dispatches with their original filesystem mtimes preserved as commit metadata; Phases 1–5 commit the test scripts, result CSVs, and figure pipeline. The pre-specified predictions are also reproduced verbatim in the docstring headers of the test scripts (e.g. `test_h_competition.py`) committed in Phases 2–4.



A **B** **C**

FIG. 1: **Apparatus validation and bulk-matched control.** (**A**) Empirical $I^{\mathrm{pop}}_{\mathrm{struct}}$ versus analytical prediction across the $\varepsilon \times L$ sweep, showing close tracking of analytical capacity in noise-free Mode 1 templating (test A1). (**B**) Separation ratio $I^{\mathrm{pop}}_{\mathrm{struct}}{}^{\mathrm{genuine}}/I^{\mathrm{pop}}_{\mathrm{struct}}{}^{\mathrm{bulk}}$ across three template-bias conditions, showing $> 20\times$ separation between genuine templating and bulk-matched controls (test A2). (**C**) Comparison-ensemble robustness: $I^{\mathrm{chan}}_{\mathrm{struct}}$ across 11 alternative comparison ensembles (test G3); the qualitative classification of Drt3a as Mode 1 and Drt3b as Mode 3 is preserved across all ensembles, confirming that the apparatus's classificatory verdict is stable under reasonable analyst choices.

TABLE I: **Substrate recursion principle: four structural conditions plus one scope-defining condition.**

Condition	Name	Statement	Failure mode in biological substrates
R1	Positive entropy per recognized site	At each addressable position, recognition alphabet $\|\Sigma\| \geq 2$ with $h_{\mathrm{site}} = \log_2 \|\Sigma\| > 0$	Mode 4 (prions, conformer-state): recognition alphabet is small attractor repertoire, h_{site} bounded at $\log_2 K$ at the macro level
R2	Unbounded extensible position count	Independently addressable non-recurrent sites: $\liminf_{L\to\infty} N(L) = \infty$	Mode 3 (cyclic): $N(L)$ bounded at cycle length N. Mode 5 (modular): bounded at module count N. Mode 4: $N(L) \approx 1$ trivially
R3	Same-operation recursion	Output is structurally readable by the same operation, possibly via reversible involution	Mode 2 (translation): protein product is not ribosome-readable; recursion is parasitic on Mode 1. Mode 5 (NRPS): peptide product is not the assembly line. Mode 6 (cortex): surface does not self-copy autonomously
R4	Heritable local drift in recognized alphabet	Drift lies in the recognition alphabet of (R1); content and templating function are the same physical entity. "Drift" rather than "error" reflects substrate-essentialism: no configuration is canonical	Crystals (excluded from biology): defect propagation in bounded geometric alphabet, not extensible alphabet. Mode 6: pattern drift independent of genome drift cannot accumulate
R5	Scope: non-teleological intrinsic operation	Operation is intrinsic physicochemistry, not externally specified; no target, loss function, or agent within the operation. Scope-defining: demarcates domain of (R1)–(R4), does not introduce a structural condition	LLM training (non-biological): operations are externally specified by loss function; gradient updates are not substrate-intrinsic recursion. Outside scope rather than structurally failing

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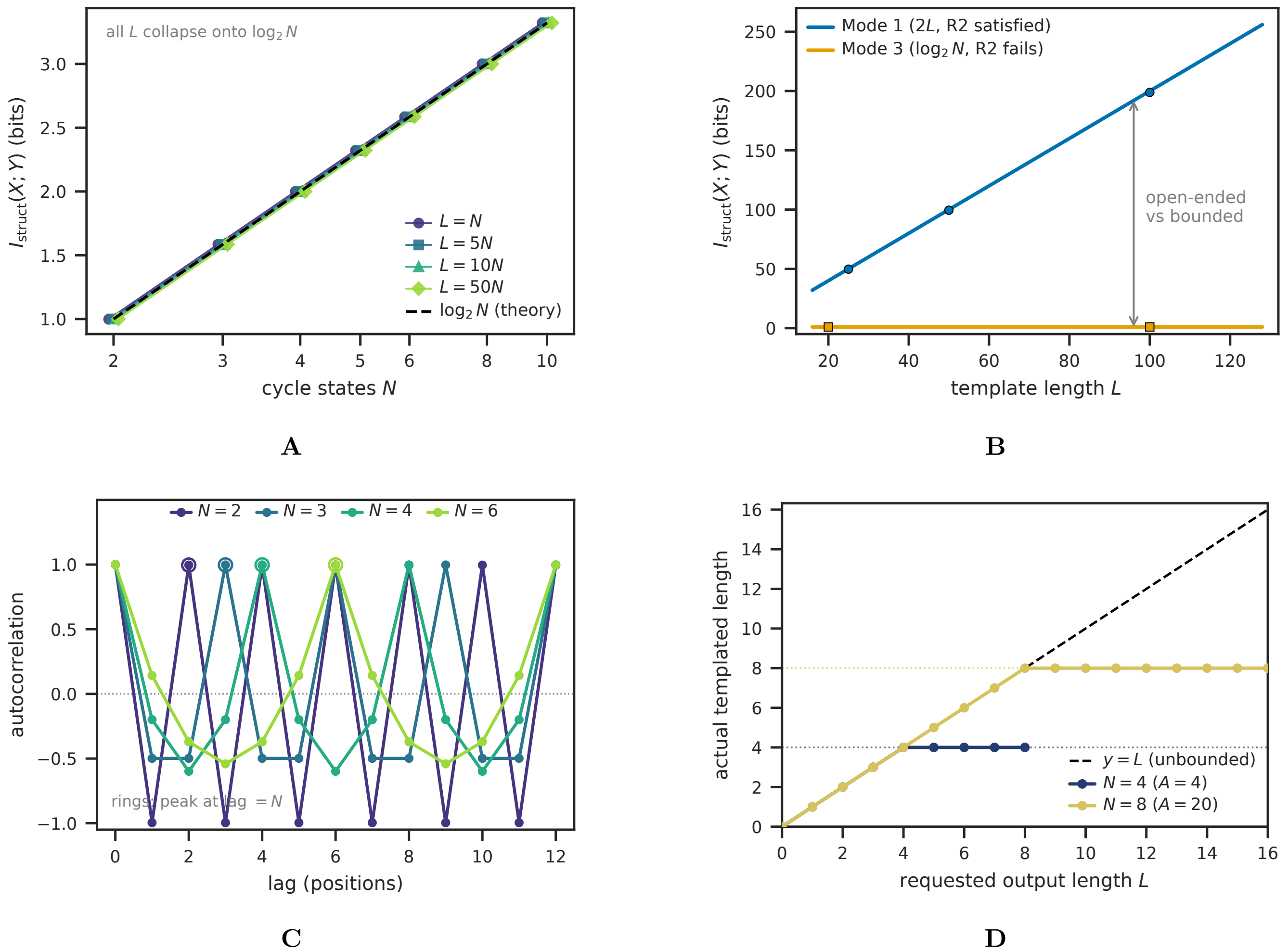

FIG. 2: **Mode-specific information-capacity scaling.** (**A**) Mode 3 information capacity $I^{\text{pop}}_{\text{struct}}$ saturates at $\log_2 N$ as cycle states N increase (test B), regardless of template length L, confirming that the macro-level capacity bound is set by the cycle's recurrent phase repertoire and not by output length. (**B**) Mode 1 capacity scales linearly in L (slope $\log_2 |\mathcal{A}| = 2$) while Mode 3 saturates at $\log_2 2 = 1$ bit; the structural difference between (R2)-satisfying and (R2)-failing substrates is visible in this single panel. (**C**) Mode 3 output autocorrelation peaks at lag $= N$ for $N \in \{2, 3, 4, 6\}$, the canonical signature of cyclic-phase templating. (**D**) Mode 5 output length is capped at the module count N regardless of requested L (test C), confirming the (R2)-failure-at-module-count diagnosis.

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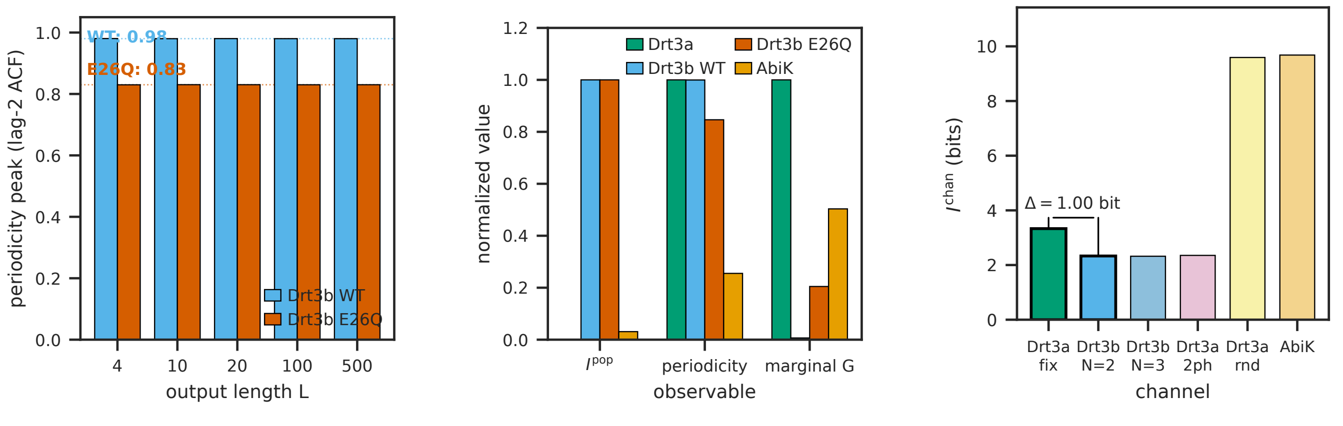

FIG. 3: **Apparatus signature on the Drt3 system.** (**A**) Drt3b WT vs E26Q periodicity peaks (test E2): WT shows $f_{\text{phase}} \approx 0.99$, E26Q shows $f_{\text{phase}} \approx 0.83$, confirming the analytical prediction of half-cycle dG misincorporation in the gate-broken mutant ($0.5 \times 0.20 = 0.10$ marginal G fraction; observed 0.1016 in Deng et al. 2026 Fig. 4J). (**B**) Apparatus triple ($I^{\text{pop}}_{\text{struct}}$, periodicity peak, marginal G fraction) separating WT Drt3b, E26Q, AbiK, and Drt3a in joint-observable space (test G2); the four systems classify cleanly despite producing nearly identical alternating output for some pairs. (**C**) Channel-mutual-information separation: against the stated five-channel comparison ensemble, Drt3a-WC channel gives $I^{\text{chan}}_{\text{struct}} = 3.33$ bits and Drt3b-N=2 channel gives $I^{\text{chan}}_{\text{struct}} = 2.33$ bits, yielding $\Delta I^{\text{chan}}_{\text{struct}} = 1.0$ bit between Mode 1 and Mode 3.

TABLE II: **Six modes of biological templating, classified by substrate recursion principle conditions.**

Mode	Substrate	Conditions failed	Capacity scaling	Copyability	Examples
1	1D linear sequence on separate molecule	None (satisfies R1–R4 within R5 scope)	$L \cdot \log_2 \lvert\mathcal{A}\rvert$	Auto-catalytic via complementarity (R3 via involution)	DNA replication, Drt3a, retrons, telomerase, DRT9
2	1D sequence + separable lookup table	R3 (parasitic recursion via Mode 1)	$\leq L \cdot H(\nu)$, code-degeneracy bounded	Via Mode 1 on encoding gene	Ribosomal translation
3	3D protein with cyclic dynamics, N states	R2 (bounded position count N)	$\leq \log_2 N$ (saturating)	Not auto-catalytically copyable	Drt3b ($N = 2$). DRT7 ($N = 1$) is structural-selectivity boundary, $\log_2 1 = 0$ positional capacity
4	Discrete conformational state	R1 (h_{site} bounded at $\log_2 K$), R2 ($N \approx 1$)	$O(1)$ length-independent	Propagation, not copy	Prions, amyloids, DRT1 filament
5	Spatial sequence of N modules	R2 (bounded at module count), R3 (parasitic)	$N \cdot \log_2 k$, output bounded by N	Via Mode 1+2 on module gene	NRPS, polyketide synthases
6	2D pattern (lattice, surface, cortex)	R3 (no autonomous self-copy), R4 (no autonomous heritable drift)	R1, R2 satisfied at surface; bounded by Mode 1 dependency	Limited heredity, parasitic on Mode 1	Quasicrystal adlayers; ciliate cortex

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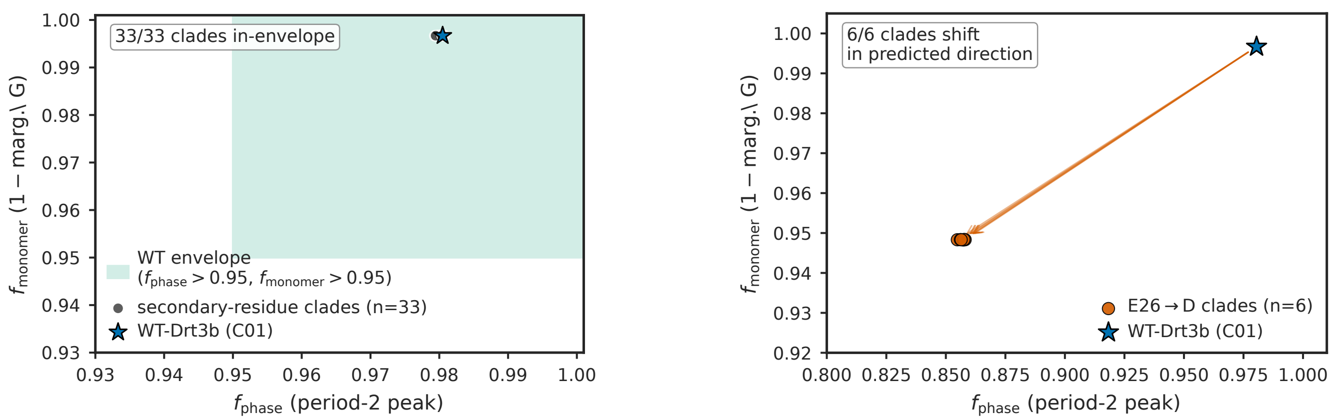


A **B**

FIG. 4: **Drt3b family-level apparatus application across 1,232 homologs.** (**A**) In-envelope behavior of 33 secondary-residue clades (test F): all 33 fall within the WT-Drt3b apparatus envelope, confirming that secondary-residue diversity does not break the cyclic-conformational mode. (**B**) E26→D selectivity-gate signature in 6 primary-gate clades (C06, C14, C20, C27, C28, C42): marginal G fraction rises from ~ 0.003 in WT-like clades to ~ 0.052 in the E26→D clades — a 17-fold elevation, consistent with residual hydrogen-bonding flexibility from the shortened Asp side chain. The periodicity peak drops to ~ 0.857 as the arithmetic consequence of A-state selectivity loss (analytical $\frac{1}{2}\sum p_A^2 + \frac{1}{2}\sum p_C^2 \approx 0.857$), not as an independent phase failure: cycle architecture is intact (R253 invariant in all 6 clades). All 6 clades show the framework-predicted selectivity shift, confirming E26 as a primary selectivity gate distinct from the architectural gate R253.

TABLE III: **DRT family classification under the framework.**

System	Mode	Mechanism summary	Reference
DRT1	Mode 4	Filamentous quaternary state with ssDNA effector	[6]
DRT2	Mode 1	Rolling-circle reverse transcription on ncRNA template, *neo* effector	[24, 25]
DRT3a	Mode 1	Watson-Crick on ACACAC ncRNA region, produces poly(GT)	[5]
DRT3b	Mode 3 ($N = 2$)	Cyclic protein-conformational gating, produces poly(AC)	[5]
DRT4	Out of scope	Random ssDNA, biasing-participant (toxin)	[26]
DRT6	Out of scope	Random ssDNA, structural homolog of DRT4	—
DRT7	Mode 3 boundary ($N = 1$)	Single-state arginine-rich pocket, $\log_2 1 = 0$ positional capacity; structural channel selectivity remains, but no positional templating; classified as boundary case rather than full Mode 3 instance	[7]
DRT9	Mode 1	Poly-U ncRNA template, produces poly(dA) homopolymer	[27]
DRT10	Mode 1	Translocation-based reuse of short ncRNA template; telomerase ancestor	[28]

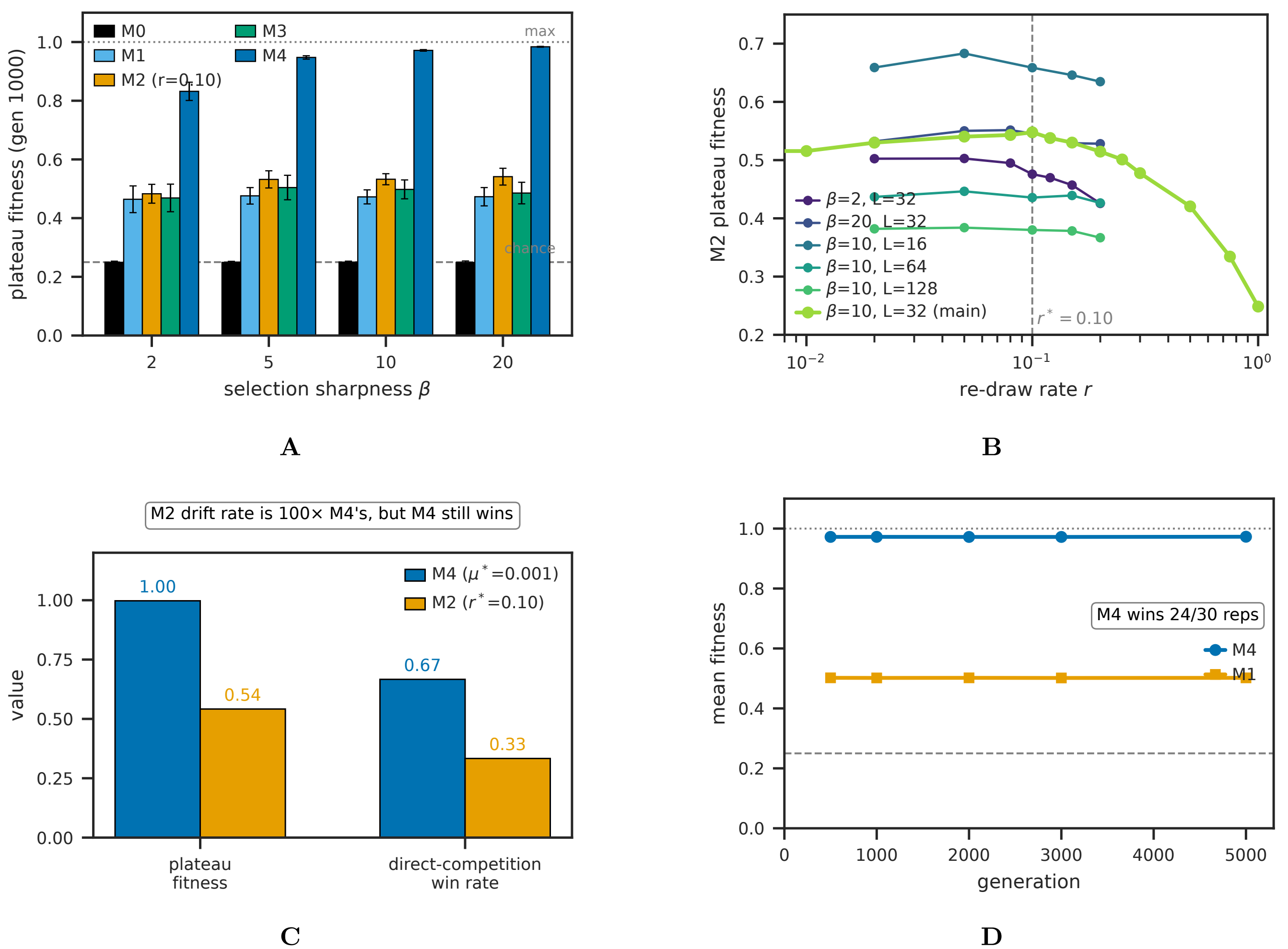
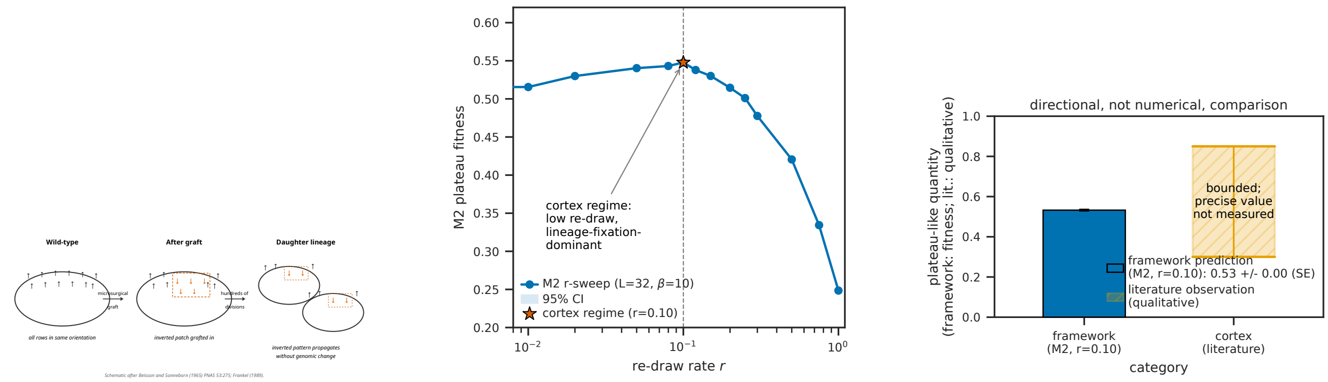


FIG. 5: **Population-dynamic plateaus from substrate condition failure.** (**A**) Mechanism-specific plateau heights $\bar{F}_{1000}$ across selection sharpness $\beta \in \{2, 5, 10, 20\}$ (test H3): M0 (R3,R4 fail) plateaus at chance (0.250); M1, M3 (R4 fails) plateau near initial-population maximum (~ 0.47–0.50); M2 at $r = 0.10$ (partial R4) plateaus at ~ 0.53; M4 (R1–R4 satisfied) reaches ~ 0.97. (**B**) M2 sweet-spot at $r^* \approx 0.10$, regime-invariant within ± 0.05 across $\beta \in \{2, 5, 10, 20\}$ and $L \in \{16, 32, 64, 128\}$ (test H5). (**C**) M4 vs M2 head-to-head at matched per-offspring drift rate (test H5): M2 introduces 2.40 new positions/offspring at $r^* = 0.10$; M4 introduces 0.02 at $\mu^* = 0.001$; M4 still wins 67%/33% in direct competition, demonstrating that targeted local drift converts to fitness gain more efficiently than wholesale lineage re-draw. (**D**) Long-horizon (5000 generations) M4 vs M1 trajectories at equal-start initial conditions ($K_{M1} = K_{M4} = 200$): M4 wins 24/30 replicates; M1's plateau is durable but bounded.



A B C

FIG. 6: **Mode 6 reframing and ciliate cortex bounded-heredity retrodiction.** (**A**) Schematic of cortical inheritance after Beisson and Sonneborn [3]: a microsurgically grafted patch of inverted ciliary basal-body rows propagates as inverted across hundreds of cell divisions in *Paramecium* without genomic change, demonstrating that the cortex inherits non-genomic structural variation. (**B**) Operational analog within the M2 mechanism class: at every cell division, cortical units are re-drawn from genome-encoded parts; the existing pattern biases the orientation of newly assembled units. The framework places the cortex in the M2 regime with $r \approx 0.10$ (low re-draw, lineage-fixation-dominant), consistent with the Beisson-Sonneborn observation. (**C**) Predicted vs observed: the framework predicts bounded but persistent heredity for Mode 6 substrates (M2 plateau at ~ 0.53 in our simulation), and the literature reports bounded but persistent cortical inheritance (hundreds of generations of pattern propagation, no qualitatively new cortical structures accumulated over evolutionary timescales). The comparison is directional, not quantitative: the literature's "bounded heredity" is a qualitative observation, not a fitness measurement.

TABLE IV: **Universal-gate site-directed-mutagenesis predictions (architectural versus selectivity).** Predictions at $L = 64$, 30 reps, $n_{\mathrm{samples}} = 5000$ per rep. 95% CIs from beta-binomial.

Mutant	Class	$I^{\mathrm{pop}}_{\mathrm{struct}}$ (bits) [95% CI]	Marginal G [95% CI]	Period-2 peak [95% CI]	Falsification target
WT (Drt3b)	—	0.9997 [0.9987, 1.0000]	0.0033 [0.0031, 0.0035]	0.9802 [0.9794, 0.9808]	—
R253A	Architectural	0.0090 [0.0065, 0.0120]	0.1265 [0.1255, 0.1274]	0.4325 [0.4303, 0.4345]	If observed $I^{\mathrm{pop}}_{\mathrm{struct}} > 0.5$ bits, R253-as-architectural-gate claim is wrong
R253K	Partial architectural	0.94	0.027	0.85	If WT-like, R253's role is more limited
R253H	Partial architectural	0.85	0.068	0.70	Intermediate; checks dose-response
G248A	Selectivity	0.9966 [0.9936, 0.9991]	0.1516 [0.1505, 0.1528]	0.7469 [0.7455, 0.7479]	If $I^{\mathrm{pop}}_{\mathrm{struct}} < 0.5$ bits, G248 is architectural and gate-class distinction fails
G248V	Selectivity	0.972	0.077	0.82	Confirms cycle preserved
G248D	Selectivity	0.971	0.102	0.79	Confirms cycle preserved
R253A + G248A	Architectural-dominant	0.0084 [0.0059, 0.0116]	0.1519 [0.1506, 0.1537]	0.3872 [0.3855, 0.3892]	If G248A signature dominates, architectural-versus-selectivity ordering is wrong

TABLE V: **Inheritance-mechanism plateau heights (Test H3, Corollary 2).** Plateau = mean fitness at gen 1000 averaged over 30 replicates. $K = 400$, $L_{\mathrm{TARGET}} = 32$, $\mu_{M4} = 0.01$. Theoretical maximum 1.0; chance baseline 0.250. Mechanism-condition mapping in column 2.

Mechanism	Conditions failed	$\beta = 2$	$\beta = 5$	$\beta = 10$	$\beta = 20$
M0 (stateless)	R3, R4	0.249	0.249	0.250	0.249
M1 (lineage fixation)	R4 (no individual-level drift)	0.464	0.476	0.472	0.473
M2 ($r = 0.10$)	R4 partial (wholesale re-draw, not local drift)	0.483	0.532	0.533	0.541
M3 (individual exact copy)	R4 (no drift)	0.469	0.504	0.498	0.485
M4 (individual + local drift)	None (R1–R4 satisfied within R5 scope)	0.832	0.948	0.972	0.984